

bioclaim

Catching fabricated and incorrect biomedical identifiers in AI-generated text

Even GPT-4o gets biomedical identifiers wrong in 1 of every 4 specialist answers — smaller models up to 68%. bioclaim catches it.

Author: Sohil Ananth Ramesh Kumar, University of Leicester · **Code:** github.com/SxR24/bioclaim · **Install:** `pip install bioclaim` · MIT-licensed

In one line. Large language models are now used across the life sciences, but they routinely cite database identifiers (UniProt, Ensembl, GO, HPO...) that look authoritative and are wrong. bioclaim is a deterministic, dependency-free tool that verifies every identifier in a piece of AI output against the authoritative source databases and flags the fabricated, mislabeled, misassigned, deprecated, and biologically-unsupported ones — with a hard guarantee that it never falsely accuses.

25–68%

OF SPECIALIST ANSWERS HAD A
WRONG IDENTIFIER (6 MODELS,
INCL. GPT-4O & GPT-5)

100%

PRECISION ON A 500-ITEM
BENCHMARK

66/66

FALSE GENE–FUNCTION CLAIMS
CAUGHT (LIVE)

1 · The problem

An LLM asked a specialist biology question returns a fluent, confident answer containing identifiers such as `UniProtKB: P04637`, `Ensembl: ENSG00000141510`, and `GO:0006915`. Any of these may be:

- **Fabricated** — a well-formed identifier that does not exist in any database.
- **Mislabeled** — a real identifier described incorrectly.
- **Misassigned** — a real, valid accession that belongs to a *different* gene.
- **Deprecated / deleted** — a real identifier that has since been retired.
- **Biologically unsupported** — a real GO term assigned to a gene that isn't annotated with it.

These errors are uniquely dangerous because they are **high-stakes** (each points, or fails to point, at a specific database record) yet **low-salience** (a short alphanumeric string a reader will not check). In a research or clinical pipeline, a single wrong accession can silently corrupt an analysis. Existing work either *measures* this (knowledge benchmarks) or builds *task-specific* self-verifying agents — but there is no general, deployable layer that can screen *any* LLM output for exactly this class of checkable error.

2 · How it works

bioclaim extracts every recognised identifier from input text and applies up to **five verification layers**, querying live authoritative databases (EBI OLS for ontologies, UniProt and Ensembl REST for genes/proteins, QuickGO for annotations). It uses **no model of its own** — it is a deterministic lookup, so it cannot itself hallucinate.

Layer	Catches	Example
1 · Format	malformed identifiers	offline, instant
2 · Existence	fabricated & deleted IDs	G0:9999999 ; deleted UniProt accessions
3 · Label	real ID, wrong description	G0:0005634 ("nucleus") called "nucleoplasm"
4 · Entity	real ID, wrong gene	P04637 given for BRCA1 (it is TP53)
5 · Function	real GO term the gene lacks	"TP53 does photosynthesis (GO:0015979)"

The core design guarantee: it never falsely accuses. Anything that cannot be verified (network failure, rate limit, ambiguous context) is marked *unverified* and is never flagged. Every lookup is cached to disk, so after a warm-up the tool is fast, offline-capable, and immune to rate limits. It ships as a Python library, a one-call API, a guardrail wrapper, and a command-line tool.

3 · The evidence — how well it solves the problem

3.1 · Six models — all wrong in 25–68% of specialist answers

Six models, including OpenAI's GPT-4o and GPT-5, were tested on 40 hard, long-tail questions (obscure genes, rare diseases, quota prompts). Every identifier was verified live, and **every flag was independently confirmed** against the source databases — zero false positives.

Model	Answers wrong	Fabricated	Mislabeled	Wrong-gene	Obsolete
Llama-3.1-8B	68% (27/40)	7	28	30	7
gpt-oss-120B	60% (24/40)	1	0	33	0
GPT-4o-mini	52% (21/40)	6	0	16	0
Llama-3.3-70B	48% (19/40)	7	5	16	1
GPT-5	40% (16/40)	0	5	10	1
GPT-4o	25% (10/40)	2	0	8	0

Even the best model (GPT-4o) is wrong 1 in 4 times — and newer is not reliably better: GPT-5, OpenAI's newest reasoning model, was *worse* than GPT-4o (40% vs 25%). Reasoning ability does not translate into factual recall of obscure identifiers. GPT-5 fabricated *zero* non-existent IDs — every error was a **real identifier used wrongly**, which is harder to detect than obvious fabrication. Asked for the disease ID for **alkaptonuria**, GPT-5 returned the ID for *systemic lupus erythematosus*; asked for **Menkes disease** it returned *Riley-Day syndrome* and fabricated supporting cross-references to appear authoritative. (A partial run of the ~1.6T DeepSeek-V4-Pro was near-zero, showing the very largest models can do better — but the relationship with capability is not monotonic.)

3.2 · Benchmark: high recall at a hard 100% precision

On 500 controlled answers with 394 injected fabrications, bioclaim was scored against ground truth. Every uncaught fabrication was a network-throttled lookup that safely returned *unverified* — **not a single fabrication was checked and missed**, and there were zero false accusations.

Metric	Value
Precision	100.0%
Recall	95.7% (377/394)
F1	0.978
False accusations	0

3.3 · Gene–function verification, validated live

On 104 curated gene–function cases run against live QuickGO, bioclaim **rejected 66/66 (100%)** implausible associations (no human protein is annotated with photosynthesis, pollen development, or nitrogen fixation) and **accepted 37/38 (97%)** genuine ones. The single exception (MTOR / "ATP binding") reflects GO annotation *incompleteness* — the tool honestly reported "not among the gene's annotations", never "false".

The most convincing demonstration

The identifier `GO:0005634` was **flagged** in one expert passage (where it was described as "nucleoplasm") and **accepted** in another (where it was correctly described as "nucleus"). The same

identifier, opposite verdicts — because the tool verifies the *claim against the database*, not the string. It also caught several genuinely **deprecated** GO terms (e.g. "cell cycle arrest", "extracellular space") that read as valid to a domain expert.

4 · The engineering rigor — problems faced & solved

bioclaim was hardened through adversarial testing on expert-authored text; each surfaced gap was diagnosed and fixed, and every flag re-verified against the live databases. This history is itself evidence of scientific care:

Problem surfaced	Resolution
Wrong-gene check false-positived on table-formatted answers	Anchored to the known gene; never guesses from proximity
Half of all UniProt accession formats were not recognised	Adopted the full official accession syntax
Deleted UniProt accessions passed as "exists" (HTTP 200)	Detect <code>Inactive</code> entries as non-resolving
An active term was mislabeled obsolete by a stale duplicate	Prefer the active, defining ontology term
GO section headers ("Cellular Component") mistaken for descriptions	Filter aspect headers from claimed labels
Annotation incompleteness could cause false accusations	Report "not among annotations" as fact, never "false"

Independent verification was performed throughout — including a case where the tool correctly flagged a term the author (and a knowledgeable reviewer) assumed was valid, and the database sided with the tool.

5 · Why it is better than the alternatives

Property	bioclaim	Knowledge benchmarks	Task-specific agents
Deployable (<code>pip install</code> , 3-line integration)	yes	no	research code
General — any identifier, any text	yes	fixed test set	single task
Deterministic (cannot itself hallucinate)	yes	n/a	often LLM-based
Verifies the <i>claim</i> (wrong-gene, gene–function)	yes	no	partial
Precision-first, never falsely accuses	yes	n/a	varies

6 · How it scales

- **Local database snapshots** — bundle GO/HPO for microsecond, fully-offline checks at corpus scale.
- **Persistent caching** (already implemented) — each identifier fetched at most once, ever.
- **Async / batch** lookups for screening large document sets.
- **Broader coverage** — RefSeq, dbSNP rsIDs, PubMed citation IDs (fabricated references), drug IDs, taxonomy.
- **Integrations** — a guardrail for LLM frameworks, a GitHub Action, a hosted API, CI checks.
- **A public benchmark** others evaluate against — defining the category.

7 · How it can be made better (roadmap)

- **Context-free detection** — catch wrong-gene and false-function claims in free text without an explicit gene hint.
- **Deeper claim verification** — gene–disease associations (ClinVar / DisGeNET), pathway membership (Reactome), variant nomenclature (HGVS/ClinVar).
- **Calibrated confidence** — a score per verdict rather than a binary flag.
- **A larger, multi-run, multi-model benchmark** — turning the finding into a citable, reference-grade dataset.
- **Peer-reviewed publication** — a manuscript draft already exists.

8 · Why this matters, and the ask

As LLMs are embedded deeper into biomedical workflows, a lightweight, deterministic safeguard for their most checkable class of error is both practical and, increasingly, necessary. bioclaim is already published, installed, tested, and validated live. The natural next steps — a rigorous multi-run benchmark, deeper claim verification, and a preprint — would benefit enormously from expert supervision.

What I'm hoping for

Your feedback on the approach and its scientific value; and, if it interests you, the opportunity to develop it further under your guidance — as a research project, a co-authored preprint, or a dissertation. A draft manuscript and a ready-to-run demo (`bioclaim < sample_paragraphs.txt` , with an answer key) are available on request.

bioclaim · open-source (MIT) · github.com/SxR24/bioclaim · `pip install bioclaim` · All figures are reproducible from the repository. Prepared by Sohil Ananth Ramesh Kumar, University of Leicester.